

GDP360 — Analyzing 50 Years of Global Growth Trends

- **Objective:** Build a KNIME-powered ETL → analysis → dashboard pipeline for GDP & macro indicators (1995–2025)
- **Audience:** Data analysts, system designers, policy researchers
- **Deliverable:** Reproducible workflows, SQL conceptual model, and interactive KPIs/dashboards
- **Visual / KNIME suggestion:** Project scope timeline (1995 → 2025)
- Data Ingest → Clean → Transform → Visualize



Introduction

What is GDP360?

360° Analytical Program

A 360° analytical program to explore GDP, population, unemployment, electricity access, and rural share across countries and years

Focus Areas

Focus: longitudinal trends, YoY growth, per-capita measures, and unemployment impact

Our Approach

Approach: SQL conceptual queries followed by KNIME implementation and visual storytelling

Refer to GDP360 Requirements Document

Datasets & Sources

Core Sources

World Data Bank (data.worldbank.org/indicator)

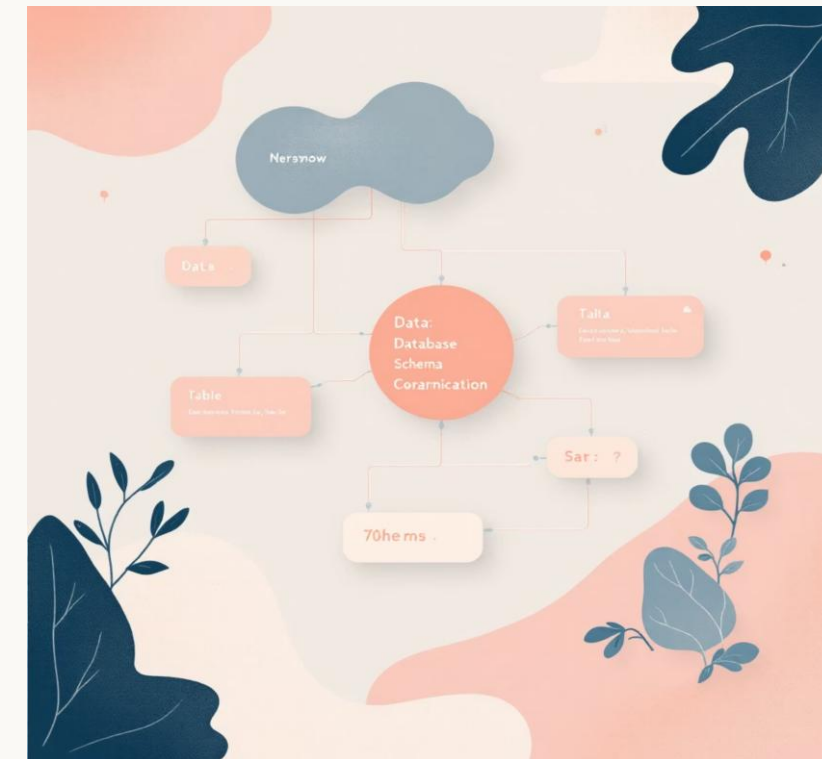
GDP (data.worldbank.org/indicator/NY.GDP.MKTP.CD)

Population (data.worldbank.org/indicator/SP.POP.TOTL)

Unemployment (data.worldbank.org/indicator/SL.UEM.TOTL.NE.ZS)

Electricity access (data.worldbank.org/indicator/EG.ELC.ACCS.ZS)

Rural Population (data.worldbank.org/indicator/SP.RUR.TOTL.ZS)



End-to-End Engineering & Data Analysis

1

Average GDP per Capita

MANDATORY

2

Actual Population of Unemployed

Actual Population of Unemployed = population × unemployment%

Additional

3

Average Unemployment %

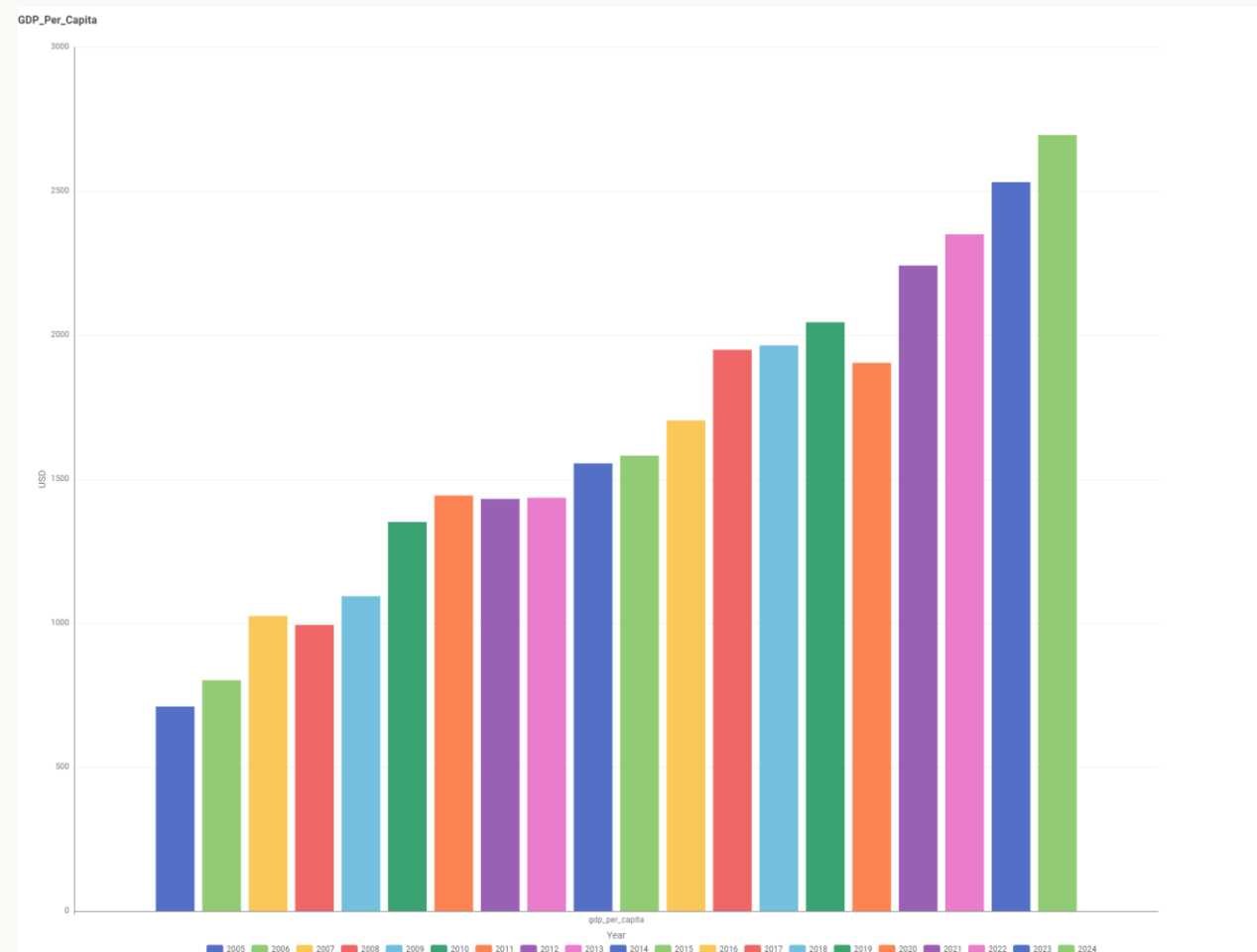
Average Unemployment % over years

Additional

Visual suggestions: line charts for trends, choropleth for absolute unemployed, KPI tiles, country/year slicers

End-to-End Engineering & Data Analysis

Average GDP per Capita **MANDATORY**



Activity 1

Database Design — Conceptual Model



Master Tables

Master tables: Country
(country_id, country_name,
country_code), Year



Fact Tables

Fact tables: GDP, Population,
UnemploymentPercent,
ElectricityPercent, RuralPopulation
— FK to Country and Year

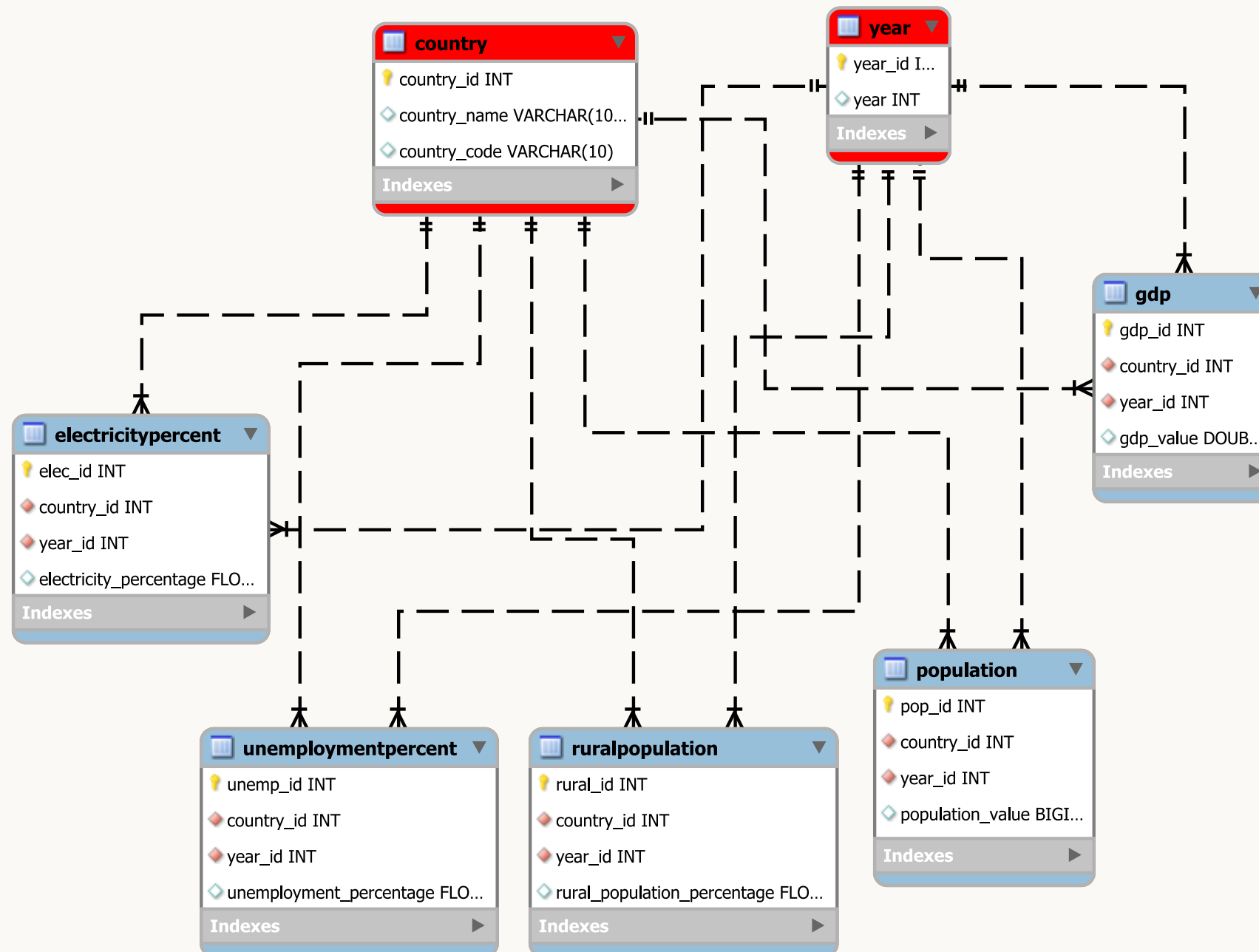


Integrity Rules

Integrity rules: referential integrity,
uniqueness on (country_id,
year_id)

Refer to DB Design Document

Database Design

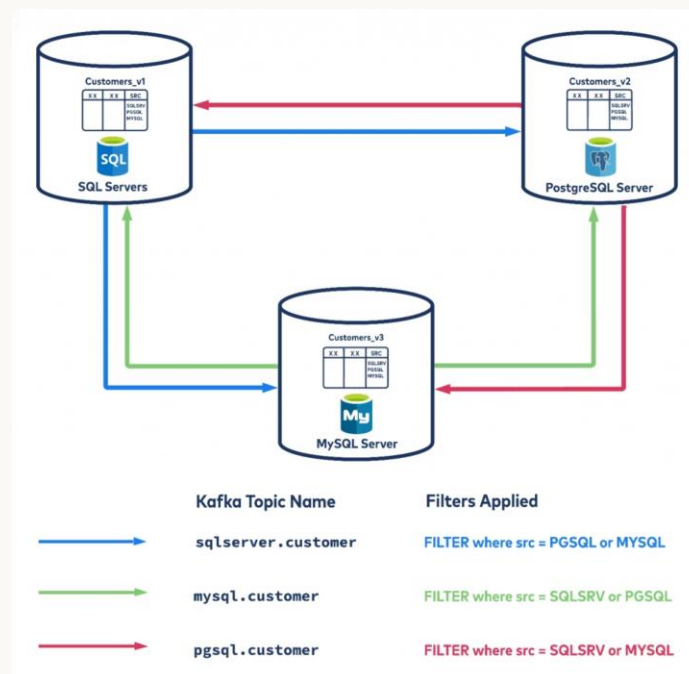


Activity 2

Importing Data: MySQL & SQLite

MySQL: Import the **gdp360.sql** file

SQLite: Import the **gdp360.db** file



Activity 3 – 6

SQL Queries

Refer to SQL Queries Document

Activity 3

Trillion-Dollar Economies

Countries with GDP > 1 Trillion

The World Economic Forum wants a quick summary of countries where the GDP value exceeds 1 trillion.

Activity 4

Top 5 GDP per Capita — 2021 Leaders

UN Population Division wants the top 5 countries by GDP per capita in 2021



Activity 5

Above-Average Electricity Access — Countries Exceeding World Yearly Average

Find all countries whose Electricity access was above the world average for the same year.



Activity 6

Rural Development Efficiency — Countries Outperforming on GDP with Low Unemployment (Yearly Report)

For a specific year, list countries where:

1. GDP is greater than the global average GDP for that year,
2. Unemployment rate is below the global average for that year,
3. Classify rural population percentage into High/Moderate/Low categories.



Activity 7A

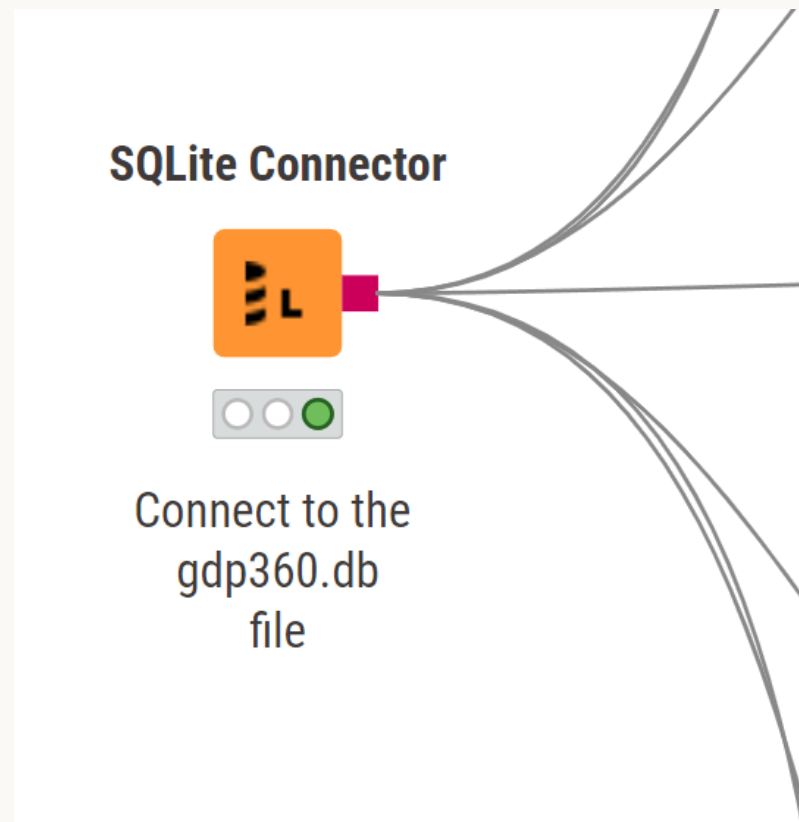
Connecting from MySQL DB to KNIME

MySQL Connector



Activity 7B

Connecting from SQLite DB to KNIME



Activity 8

Removing Duplicate Country Names and Codes

Problem

Duplicate country names/codes cause join duplication

Approach

Identify duplicates by country_code and Names

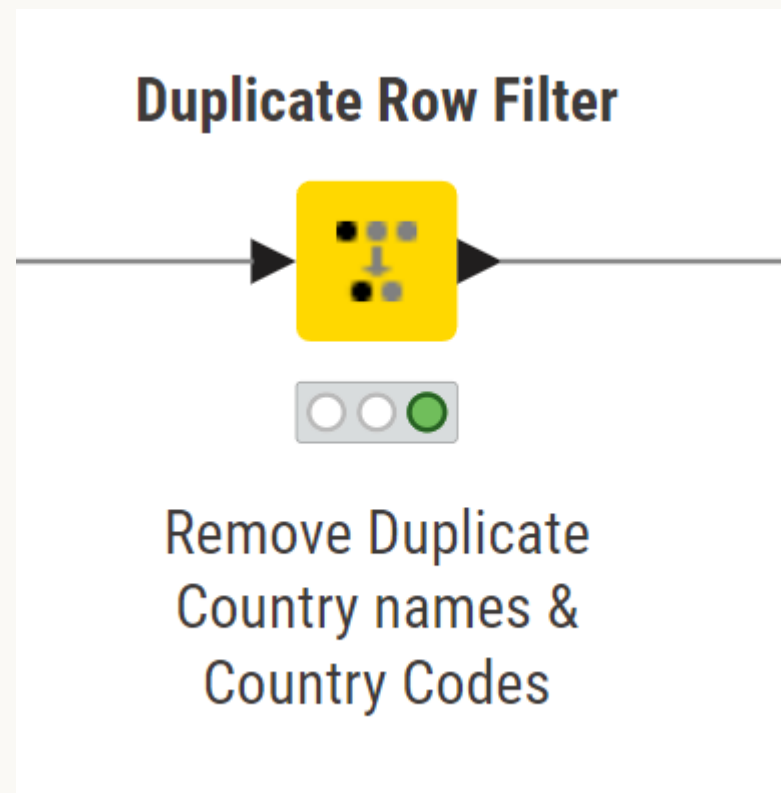
KNIME nodes

String Manipulation

Duplicate Row Filter

Activity 8

Removing Duplicate Country Names and Codes



Activity 9

Removing Missing Country Name or Codes

Problem

Null country identifiers break FK joins

Activity 9

Removing Missing Country Name or Codes



Activity 10

Removing Future Years Greater than 2025

Problem

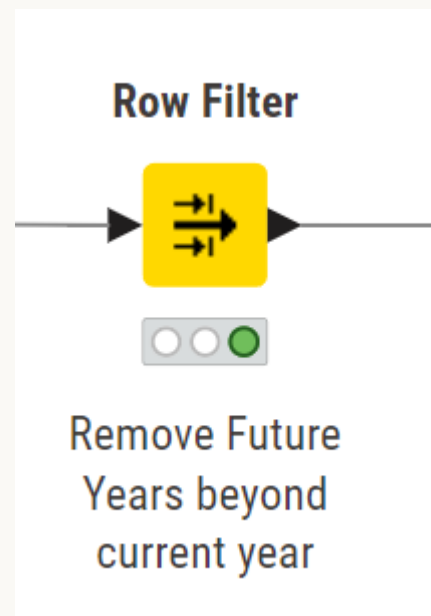
Future years outside project scope (post-2025)

Approach

Filter records where year > 2025 into exception table instead of deleting raw data

Activity 10

Removing Future Years Greater than 2025



Activity 11

Remove unemployment% > 100%



Problem

Unemployment percentage > 100 indicates bad data or different units



Approach

Remove Rows

Activity 11

Remove unemployment% > 100%

